M2: Improved DeepLabV3+: all results tables

J. Zhang et al., Sensors [P01]; re-implemented (no code released) · generated from the saved metric files

Fixed order in every table. Models: (1) trained on the **LEAKY** official SOS split, (2) trained on the **LEAK-FREE** group-disjoint split. Tests: SOS official test → SOS leak-free test → Peru all → Peru test subset → MDOD. Metrics: Acc, P, R, F1, IoU oil, IoU bg, mIoU, BF1, F1 official script, ROC-AUC, PR-AUC, ECE, MSE, R^2 (percentages for the first nine). Peru and MDOD were never used for training.

Metric definitions (same order as every table)

Table 1. Metrics. TP / FP / FN / TN = oil pixels predicted correctly / false oil / missed oil / correct background. Pixels are pooled over all test tiles. Higher is better except ECE and MSE.

Metric	Short	What it measures	Formula	Good value
Accuracy	Acc	share of all pixels classified correctly (inflated by the large sea background)	(TP+TN)/all	high, but not informative alone
Precision (oil)	P	of the pixels called oil, how many are oil (false-alarm control)	TP/(TP+FP)	high
Recall (oil)	R	of the real oil pixels, how many were found (missed-oil control)	TP/(TP+FN)	high
F1 (oil)	F1	balance of precision and recall; equals the Dice coefficient	2PR/(P+R)	high
IoU (oil)	IoU oil	overlap of predicted and true oil areas; stricter than F1	TP/(TP+FP+FN)	high
IoU (background)	IoU bg	same for the background class	TN/(TN+FP+FN)	high
mIoU	mIoU	mean of oil IoU and background IoU; the main SOS benchmark number. Rises when the test set has more background	(IoU oil + IoU bg)/2	high
Boundary F1	BF1	F1 of the slick outlines, matched within 2 pixels	—	high
F1, official script	F1 off.	what the official SOS script reports: F1 of the BACKGROUND class (a bug in that script)	2·Pbg·Rbg/(Pbg+Rbg)	for comparison only
ROC-AUC	ROC-AUC	how well probabilities rank oil pixels above non-oil, at any threshold	area under ROC	1.0 (0.5 = chance)
PR-AUC	PR-AUC	same, focused on the oil class; sensitive to class imbalance	area under precision-recall	high
ECE	ECE	calibration: gap between predicted probability and how often it is right (15 bins)	$\Sigma\ (n_b/N)\lvert acc_b - conf_b\rvert$	0
MSE (= Brier score)	MSE	mean squared error between predicted oil probability p and true label y (0/1) of every pixel	mean $(p - y)^2$	0
R^2	R^2	share of the label variance explained by the probabilities; 0 = no better than predicting the average oil share, negative = worse	$1 - MSE / (f(1 - f))$, f = oil share	1
TTA		test-time augmentation: average of 8 rotated / flipped predictions		
Alert rate (MDOD)		share of boxes in which at least x % of pixels have p(oil) > 0.5		high for oil, low for look-alikes
AUC oil vs class (MDOD)		chance that a random oil box scores higher than a random box of that class	Mann-Whitney	1.0 (0.5 = chance)

1. Master table: every test, every metric

Table 2. All pooled metrics (%, except ROC-AUC, PR-AUC, ECE, MSE and R^2). The SOS leaky-split model is tested on the official test set, the leak-free model on the leak-free test set (each model's own test set).

Test	Model	Mode	Acc	P	R	F1	IoU oil	IoU bg	mIoU	BF1	F1 off.	ROC-AUC	PR-AUC	ECE	MSE	R^2
SOS official test (839 tiles)	leaky-trained	TTA	91.56	86.64	89.50	88.05	78.65	87.76	83.20	41.12	93.48	0.9720	0.9557	0.0316	0.0632	0.721
SOS official test (839 tiles)	leaky-trained	single	91.43	86.36	89.43	87.87	78.36	87.57	82.97	40.64	93.37	0.9711	0.9546	0.0357	0.0647	0.715
SOS leak-free test (835 tiles)	leak-free-trained	TTA	92.49	88.03	86.42	87.22	77.33	89.91	83.62	47.53	94.68	0.9771	0.9524	0.0241	0.0554	0.734
SOS leak-free test (835 tiles)	leak-free-trained	single	92.34	87.78	86.13	86.95	76.91	89.71	83.31	46.93	94.57	0.9760	0.9500	0.0279	0.0572	0.726
Peru, all 2,013 patches	leaky-trained	TTA	60.21	13.01	94.84	22.89	12.92	57.72	35.32	2.66	73.19	0.9080	0.4170	0.2955	0.3211	-4.500
Peru, all 2,013 patches	leak-free-trained	TTA	59.17	12.66	94.23	22.32	12.56	56.62	34.59	2.78	72.30	0.9005	0.4121	0.3002	0.3295	-4.644
Peru test subset (338 patches)	leaky-trained	TTA	66.53	9.28	97.36	16.94	9.25	65.35	37.30	1.98	79.04	0.9356	0.3645	0.2456	0.2714	-7.025
Peru test subset (338 patches)	leak-free-trained	TTA	66.55	9.19	96.22	16.78	9.16	65.38	37.27	2.53	79.07	0.9130	0.2792	0.2433	0.2760	-7.161
SOS official test: published	Zhang et al. [P01]					89.62			81.34							

MSE and R^2. MSE is the mean squared error between the predicted oil probability and the true 0/1 label of every pixel (the Brier score). $R^2 = 1 - \text{MSE} / \text{Var(label)}$, where $\text{Var(label)} = f(1 - f)$ and f is the oil share of the test pixels. $R^2 = 1$ is perfect, 0 equals always predicting the average oil share, and a negative value is worse than that. These are regression metrics, so they are computed on the probabilities, not on the thresholded masks.

2. Reproduction check (SOS official test, TTA)

Table 3. Ours vs published (Zhang et al. [P01]).

Metric	Ours	Published	Gap (pp)	Within ±2 pp
mIoU	83.20	81.34	+1.86	yes
F1 (oil)	88.05	89.62	-1.57	yes

3. Leaky vs leak-free SOS split (TTA)

Table 4. Same model and recipe; only the SOS split changes.

Metric	Leaky split (official test)	Leak-free split (leak-free test)	Difference (leak-free – leaky)
Accuracy	91.56	92.49	+0.93 pp
Precision (oil)	86.64	88.03	+1.39 pp
Recall (oil)	89.50	86.42	-3.08 pp
F1 (oil)	88.05	87.22	-0.83 pp
IoU (oil)	78.65	77.33	-1.32 pp
IoU (background)	87.76	89.91	+2.15 pp
mIoU	83.20	83.62	+0.41 pp
Boundary F1	41.12	47.53	+6.41 pp
F1, official script (background class)	93.48	94.68	+1.20 pp
ROC-AUC	0.9720	0.9771	+0.0051
PR-AUC	0.9557	0.9524	-0.0033
ECE	0.0316	0.0241	-0.0075
MSE (= Brier score)	0.0632	0.0554	-0.0078
R^2	0.721	0.734	+0.0131
Oil share of test pixels	34.72	29.64	-5.08 pp
Test tiles	839	835	

4. Training details and per-tile distribution

Table 5. Training and per-tile oil IoU (TTA; tiles that contain oil).

Model	Training tiles	Epochs	Training time	Evaluated checkpoint	Its training loss	Final-epoch loss	Parameters	Oil test tiles	Per-tile IoU median	Tiles with IoU < 0.3
Trained on LEAKY official split	3354	80	1 h 10 min	final epoch	0.3098	0.3098	6.04 M	805	0.701	4.1 %
Trained on LEAK-FREE group split	3358	80	1 h 12 min	final epoch	0.3101	0.3101	6.04 M	803	0.689	4.2 %

Setup: Adam 1e-3, batch 15, 80 epochs, cosine decay (assumed); flips + 90° rotations (assumed); BCE + Dice; final-epoch checkpoint.

5. Overfitting check: training tiles vs test tiles (single pass, no augmentation)

Table 6. Each model scored on its own training tiles and its test tiles. A large positive gap means overfitting.

Model	Data	Acc	P	R	F1	IoU oil	IoU bg	mIoU	BF1	ROC-AUC	ECE	MSE	Loss (BCE+Dice)
Trained on LEAKY official split	train (3354 tiles)	93.59	89.57	88.43	89.00	80.17	91.35	85.76	50.43	0.9833	0.0199	0.0472	0.3160
Trained on LEAKY official split	test (839 tiles)	91.43	86.36	89.43	87.87	78.36	87.57	82.97	40.64	0.9711	0.0357	0.0647	0.3842
	gap (train − test)	+2.17	+3.21	-1.00	+1.13	+1.81	+3.79	+2.80	+9.79	+0.0121	-0.0158	-0.0175	-0.0682
Trained on LEAK-FREE group split	train (3358 tiles)	93.50	90.26	88.28	89.26	80.60	91.10	85.85	48.83	0.9830	0.0190	0.0478	0.3135
Trained on LEAK-FREE group split	test (835 tiles)	92.34	87.78	86.13	86.95	76.91	89.71	83.31	46.93	0.9760	0.0279	0.0572	0.3621
	gap (train − test)	+1.17	+2.48	+2.15	+2.31	+3.69	+1.40	+2.54	+1.91	+0.0070	-0.0089	-0.0094	-0.0486

6. 95 % confidence intervals (bootstrap)

Table 7. Value [2.5 %, 97.5 %] over 2,000 resamples (MDOD: 500). SOS resamples tiles; Peru resamples whole scenes (40), because tiles of one scene are not independent. TTA predictions.

Test	Model	F1 (oil)	IoU (oil)	mIoU	Precision	Recall	Look-alike called oil	MDOD AUC oil vs look-alikes
SOS (own test set), by tile	Trained on LEAKY official split	88.05 [86.93, 89.04]	78.65 [76.89, 80.24]	83.20 [82.01, 84.29]	86.64 [85.14, 87.97]	89.50 [88.04, 90.81]	—	—
SOS (own test set), by tile	Trained on LEAK-FREE group split	87.22 [86.10, 88.17]	77.33 [75.59, 78.84]	83.62 [82.57, 84.61]	88.03 [86.71, 89.21]	86.42 [84.97, 87.69]	—	—
Peru all (by scene) / MDOD (by box)	Trained on LEAKY official split	22.91 [9.45, 34.34]	12.94 [4.96, 20.73]	35.36 [30.25, 40.25]	13.03 [4.97, 20.96]	94.82 [91.62, 96.02]	87.38 [82.16, 92.04]	0.293 [0.259, 0.327]
Peru all (by scene) / MDOD (by box)	Trained on LEAK-FREE group split	22.34 [8.99, 34.52]	12.58 [4.71, 20.86]	34.63 [29.10, 39.69]	12.67 [4.73, 21.10]	94.21 [91.65, 96.41]	90.37 [85.56, 94.96]	0.286 [0.252, 0.319]

Table 8. Paired comparison on the same Peru patches (scene bootstrap).

Peru metric	Leaky-trained	Leak-free-trained	Difference (leak-free − leaky) [95 % CI]	Clear difference?
F1 (oil)	22.91	22.34	-0.57 [-1.29, +0.07] pp	no (CI includes 0)
IoU (oil)	12.94	12.58	-0.36 [-0.87, +0.04] pp	no (CI includes 0)
Precision (oil)	13.03	12.67	-0.36 [-0.83, +0.05] pp	no (CI includes 0)
Recall (oil)	94.82	94.21	-0.62 [-1.78, +1.09] pp	no (CI includes 0)

7. Peru: what is called oil, per true class (TTA)

Table 9. Share of each true Peru class predicted as oil (%). For "oil" this is the recall.

True class	Share of pixels (all)	Called oil: leaky-trained (all)	Called oil: leak-free-trained (all)	Share of pixels (test subset)	Called oil: leaky-trained (test subset)	Called oil: leak-free-trained (test subset)
oil	6.2	94.8	94.2	3.5	97.4	96.2
look-alike	5.3	87.4	90.4	8.4	84.2	86.4
ship	0.1	61.0	63.9	0.0	90.7	90.0
sea	84.1	40.7	41.8	85.7	30.5	30.2
land	4.3	13.5	11.0	2.4	6.9	5.9

8. MDOD: box-level alerts (single pass, 400 boxes per class)

Table 10. Alert @x % = share of boxes with at least x % of pixels at p(oil) > 0.5 (%). For oil spills high is good; for look-alikes low is good. AUC below 0.5 = look-alikes score higher than oil.

Class	Model	Boxes	Alert @2 %	Alert @10 %	Alert @20 %	Alert @50 %	Mean p(oil)	Median coverage	Mean peak p(oil)	AUC oil vs class (mean)	AUC oil vs class (peak)
oil spill	leaky-trained	400	57.0	46.0	36.5	20.0	0.253	0.070	0.596	—	—
	leak-free-trained	400	50.5	40.0	32.8	16.5	0.221	0.028	0.515	—	—
biogenic slick	leaky-trained	400	96.5	86.5	70.0	19.8	0.335	0.288	0.968	0.334	0.100
	leak-free-trained	400	94.5	82.2	60.2	12.5	0.287	0.235	0.956	0.330	0.101
low wind	leaky-trained	400	99.2	99.0	98.2	71.8	0.624	0.629	0.992	0.168	0.050
	leak-free-trained	400	99.0	98.8	97.0	63.0	0.581	0.574	0.989	0.159	0.053
rain cell	leaky-trained	400	91.0	74.5	54.8	20.8	0.293	0.223	0.906	0.387	0.220
	leak-free-trained	400	86.5	69.0	49.0	16.0	0.262	0.190	0.875	0.373	0.214
sea ice	leaky-trained	400	94.8	88.5	78.0	37.8	0.435	0.390	0.953	0.284	0.121
	leak-free-trained	400	92.5	86.2	72.2	30.8	0.394	0.333	0.936	0.279	0.126
all look-alikes	leaky-trained	1600								0.293	0.123
all look-alikes	leak-free-trained	1600								0.286	0.124

9. Where the method is weak (critique) and what a contribution must beat

Each row is a metric on which the method performs badly. **Baseline for contributions = the leak-free-trained model** (the fair one). A contribution must beat it by at least 1 pp (0.01 for PR-AUC, ECE, MSE, R^2 and AUC). SOS rows use each model's own test set. Single seed, so differences under about 1 pp on SOS are within noise (see section 6).

Table 11. Weak points (%, except PR-AUC, ECE, MSE, R^2 and AUC). Computed from the saved predictions; no model was re-run.

Metric	Better	Leaky-trained	Leak-free-trained	Target (beat leak-free by ≥ 1 pp)	Why it is weak
SOS oil IoU (each model's own test set)	higher	78.65	77.33	> 78.33	Headline in-domain number; the leak-free value is the honest one.
SOS oil recall	higher	89.50	86.42	> 87.42	Missed oil; falls when the leak is removed.
SOS boundary F1	higher	41.12	47.53	> 48.53	Slick outlines are poor (partly coarse polygon labels).
SOS small-spill tiles: mean IoU	higher	37.42	30.81	> 31.81	Thin / small slicks are the hardest tiles.
SOS oil tiles with IoU < 0.3	lower	4.10	4.23	< 3.23	Share of tiles that are essentially failed.
Peru precision (oil)	higher	13.01	12.66	> 13.66	About 6 false-alarm pixels per true oil pixel on a new sea.
Peru F1 (oil)	higher	22.89	22.32	> 23.32	Transfer to a new sea collapses (≈ 87 % → 22–29 %).
Peru IoU (oil)	higher	12.92	12.56	> 13.56	Same failure, stricter metric.
Peru look-alike pixels called oil	lower	87.42	90.41	< 89.41	The core failure: SOS has no look-alike class.
Peru PR-AUC	higher	0.417	0.412	> 0.422	Ranking of oil vs everything else on the new sea.
Peru ECE (calibration)	lower	0.295	0.300	< 0.290	Probabilities are over-confident (≈ 0.03 on SOS).
Peru MSE (= Brier score)	lower	0.321	0.329	< 0.319	Probability error, combines calibration and accuracy.
Peru R^2 (probabilities vs labels)	higher	-4.500	-4.644	> -4.634	Negative: worse than predicting the average oil share everywhere.
MDOD oil boxes alerted (@ 2 %)	higher	57.00	50.50	> 51.50	Many real oil boxes are missed at MDOD's coarser resolution.
MDOD look-alike boxes alerted (@ 2 %, mean of 4 classes)	lower	95.38	93.12	< 92.12	Almost every look-alike raises an alarm.
MDOD AUC, oil vs look-alikes	higher	0.293	0.286	> 0.296	Below 0.5: look-alikes score higher than oil.

Table 12. SOS per-tile oil IoU by spill size (share of the 256 × 256 tile covered by true oil; TTA).

Spill size in tile	Model	Oil tiles	Mean IoU	Median IoU	Tiles with IoU < 0.3
small spill (< 2 % of tile)	leaky-trained	38	37.42	39.20	34.21
	leak-free-trained	42	30.81	30.16	50.00
medium (2–20 %)	leaky-trained	299	55.42	54.75	4.68
	leak-free-trained	332	57.77	57.01	3.01
large (≥ 20 %)	leaky-trained	468	79.39	82.94	1.28
	leak-free-trained	429	77.88	78.89	0.70

Oil-free test tiles: 34 (official test) and 32 (leak-free test). Share with a false alarm covering at least 1 % of the tile: 0.00 % and 0.00 %, so clean SOS sea is not a weakness; the false alarms appear on look-alikes in other seas.